Soil Carbon Estimation From Hyperspectral Imagery With Wavelet Decomposition and Frame Theory

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***Abstract*— Assessing soil organic carbon (SOC) stocks is crucial for understanding the carbon sequestration potential of agroecosystems and for mitigating climate change. This study presents a novel method for assessing SOC and mineral content at various soil depths in sorghum crops using hyperspectral remote sensing. Conducted at Planthaven Farms, MO, the research encompassed ten genotypes across 30 plots, yielding 180 soil samples from six depth intervals (0–150 cm) of bare soil. Chemical analyses determined the SOC and mineral levels, which were then compared to spectral data from HySpex indoor sensors. We utilized time-frequency analysis methods, including discrete wavelet transformation (DWT), continuous wavelet transformation (CWT), and frame transformation along with traditional spectral transformations, specifically fractional derivatives and continuum removal. The analysis revealed the shortwave infrared (SWIR) region, particularly the 1800–2000 nm range, as having the strongest correlations with SOC content (with R^2 exceeding 0.8). The visible near-infrared (VNIR) region also provided valuable insights. Models incorporating CWT achieved high accuracy (test R^2 exceeding 0.9), while frame transformation achieved strong accuracy (test R^2 between 0.7 and 0.8) with fewer features. The random forest regressor (RFR) proved to be most robust, demonstrating superior accuracy and reduced overfitting compared to support vector regression (SVR), partial least squares regression (PLSR), and deep neural network (DNN) models. The models demonstrated the efficacy of hyperspectral data for SOC estimation, suggesting potential for future applications that integrate this data with above-ground biomass to improve SOC mapping across larger scales. This research offers a promising spectral transformation approach for effective carbon management and sustainable agriculture in a changing climate.**

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I. Introduction

AGROECOSYSTEMS, which account for one-third of the global arable land, are vital in the global carbon cycle, serving as significant carbon sinks by sequestering large amounts of organic carbon in the soil [1]. This process is critical for mitigating climate change by reducing atmospheric carbon dioxide levels [2]. Soil organic carbon (SOC) is a key component of soil health, enhancing water retention, nutrient availability, soil biodiversity, and erosion resistance, thereby supporting optimal crop productivity. SOC is comprised of carbon found in living, dead, and decomposing organic materials within the soil, including plant material, soil organisms, and animal remains [3]. The role of SOC in improving the physical, chemical, and biological properties of soil is well documented, making it essential for fostering optimal crop productivity [4].

Accurately quantifying SOC is inherently challenging due to soil's high spatial variability and the complexities introduced by differing sampling techniques, such as variations in depth and analysis increments [5]. Traditional field methods are labor-intensive and primarily focus on topsoil layers, often overlooking the substantial carbon stocks in deeper soils, which account for nearly half of the global SOC reserves [6].

The advent of hyperspectral imaging has significantly advanced soil analysis by capturing continuous spectra for each pixel, allowing for the identification of distinct spectral signatures associated with SOC [7]. In its developmental stages, hyperspectral imaging was utilized primarily for identifying minerals, rocks, and soils through handheld reflectance spectroscopy, with its applications expanding significantly over time [8]. Despite these advancements, the application of hyperspectral imaging in depth-sensitive SOC estimation remains limited. Recent developments in preprocessing methods, such as Savitzky–Golay derivatives and wavelet decomposition, have improved SOC prediction accuracy by enhancing the signal-to-noise ratio and extracting valuable spectral information [9]. However, these improvements have not fully

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addressed the complexities of SOC distribution across different soil layers.

Recent advancements in AI techniques, such as SpectralGPT, have rapidly developed and are increasingly applied in hyperspectral remote sensing for SOC estimation [10]. However, harnessing the power of hyperspectral data requires robust models [11] that can extract meaningful features from across hundreds to thousands of bands. Integrating multifeature fusion approaches, like combining graph neural networks with CNNs, enhances hyperspectral image classification [12]. While these methods offer future potential, this study utilizes traditional machine learning to identify key spectral regions for SOC detection through signal transformation methods. These foundational insights can be utilized in future studies on a larger scale, with more diverse soil types and geographical coverage, potentially integrating advanced AI models for enhanced prediction accuracy. Furthermore, research underscores the critical role of precise variable selection in enhancing the accuracy of SOC estimation models, as variable selection directly influences model performance by filtering out irrelevant data [13]. The challenges associated with transferring visible and near-infrared (VNIR) models across different environmental contexts further emphasize the need for adapting models to the specific spectral and soil conditions of diverse study areas [14], [15].

Wavelet decomposition and frame theory offer innovative approaches to address these challenges. Wavelet decomposition excels in capturing both low and high-frequency components of hyperspectral signals, facilitating tasks like feature extraction and denoising, while frame theory provides a robust framework for processing complex, high-dimensional datasets [16]. Frame theory, with its foundation in linear algebra and signal processing, provides a comprehensive framework for analyzing complex, high-dimensional datasets, ensuring perfect reconstruction, stability, redundancy, and efficient, signal-independent linear inversion procedures [17]. The theoretical principles of frame theory, which are essential for processing and interpreting high-dimensional datasets in hyperspectral imagery analysis, are well articulated and supported by the foundational work of Kovačević & Chebira [18]. This attribute is crucial for analyzing intricate or nonstationary characteristics of hyperspectral data, thereby enhancing the accuracy of SOC estimations [19].

Enhanced SOC quantification methods can improve carbon management strategies, aiding climate change mitigation through accurate monitoring of carbon sequestration in agroecosystems [20]. By providing detailed SOC assessments across different soil depths, this research supports innovative soil management approaches, contributing to the resilience and sustainability of agricultural systems worldwide [21]. The objectives of the work presented in this article are to: 1) identify spectral regions most sensitive to SOC changes and the influence of soil depth; 2) quantify the improvement of wavelet decomposition and frame theory over traditional hyperspectral analysis methods; and 3) develop an optimized SOC estimation model incorporating wavelet decomposition and frame theory.

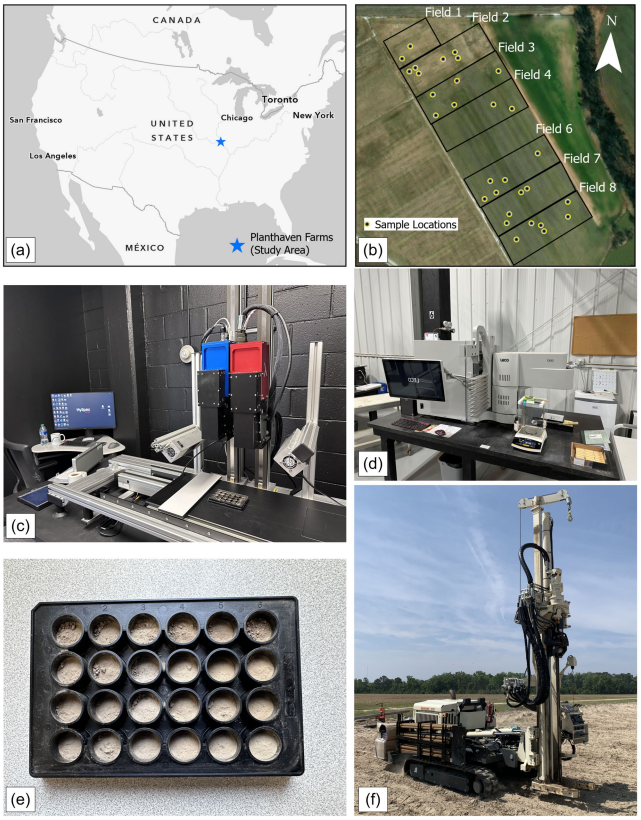


Fig. 1. Location of the study area and experimental setup. (a) Geographic location of the study area. (b) Location of the seven test fields. (c) Indoor hyperspectral scanner. (d) LECO 832 analyzer. (e) Processed soil samples in the tray. (f) GeoProbe machine.

II. Data

A. Field Data Collection

The field experiment was conducted at Planthaven Farms, MO, encompassing seven fields strategically positioned near the Missouri River watershed [Fig. 1(a) and (b)]. These fields, totaling 2.2 hectares, were arranged in a randomized complete block design. We planted 308 sorghum plots, including ten select genotypes replicated three times for comprehensive analysis. Plot dimensions were standardized at 20 × 20 ft, with eight rows of sorghum per plot. Planting occurred in late June, and baseline soil samples were collected prior to planting. Soil samples were collected using a GeoProbe machine [Fig. 1(f)] to extract cylindrical cores of 150 cm depth and 7.6 cm diameter. These cores were stored at temperatures between 0 °C and 4 °C to preserve their integrity until further processing.

B. Laboratory Analysis

Soil core processing involved several steps (Fig. 2). The polyvinyl chloride (PVC)-encased cores were sectioned using a bandsaw into six distinct depth intervals: 0–15, 15–30, 30–60, 60–90, 90–120, and 120–150 cm. Two cups of soil were reserved from each depth in a plastic bag, then dried at 40 °C–45 °C until completely devoid of moisture and subsequently stored for analysis. We selected Ward Laboratories Inc. (Kearney, NE) for our carbon analysis needs. Their LECO

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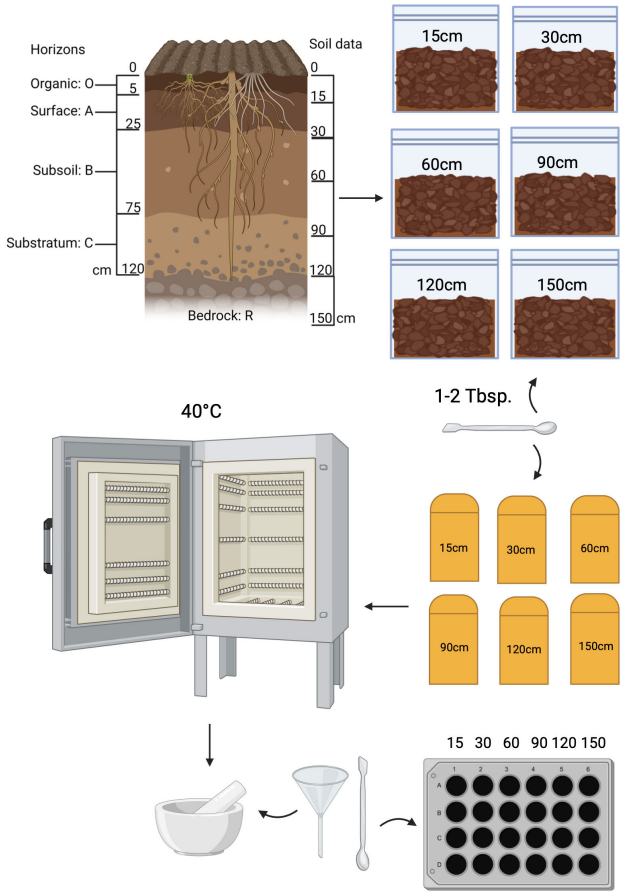


Fig. 2. Soil sample processing workflow.

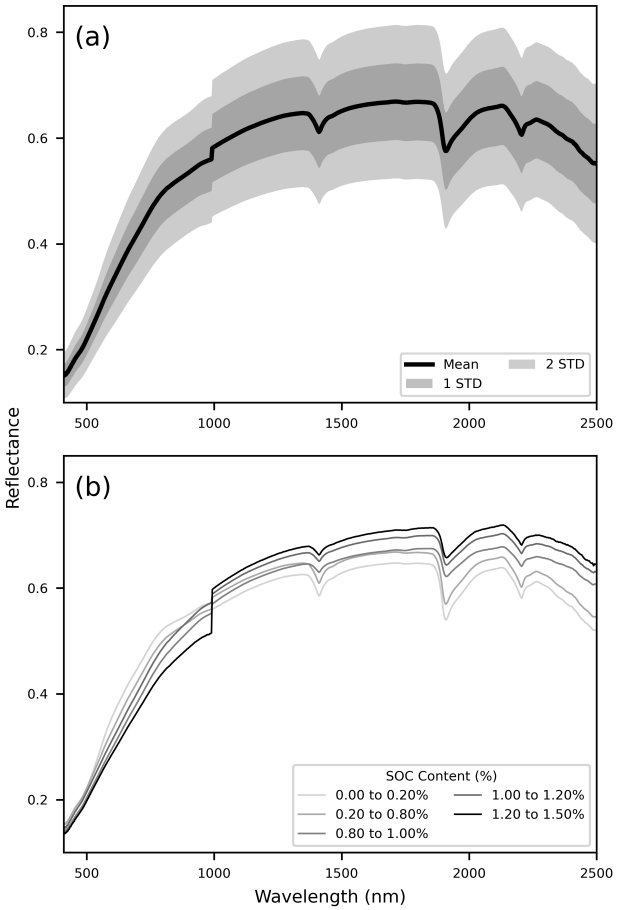


Fig. 3. Exploratory results. (a) Mean, 1 and 2 standard deviations of the sample spectra ($n = 180$). (b) Reflectance spectra for soils with varying SOC content.

832 instrument [Fig. 1(d)], operating on a combustion-based principle, provided precise SOC quantification.

C. Hyperspectral Imagery

To complement soil carbon analysis, we utilized advanced indoor hyperspectral scanning of dried soil samples using the HySpex sensor system, which captures both VNIR and short-wave infrared (SWIR) regions. The VNIR sensor covers 400–1000 nm across 300 bands with 1.8 nm spectral resolution, while the SWIR sensor spans 960–2500 nm across 362 bands with 1.2 nm spectral resolution. The native spatial resolution of VNIR and SWIR systems are 0.03 and 0.15 mm/pixel for VNIR and SWIR regions, respectively. Custom close-up lenses were used to maintain a 30 cm working distance, and the setup included vertical sensor mounting with halogen lamps at 45° angles to minimize noise. A 50% zenith white reflectance panel corrected the raw data, and the HySpex RAD module facilitated radiometric calibration, converting raw values into absolute radiance (W/sr·nm·m^2) [22]. Preprocessed samples were organized in 24-cell assay plates, with each row representing six soil depths from a plot for systematic scanning. Fig. 3(a) and (b) illustrate the mean and variability of the samples and the spectra's relationship with SOC content, while Fig. 4 shows the decline in SOC with increasing depth. Table I details the dataset's central tendency and spread.

TABLE I
DESCRIPTIVE STATISTICS OF SOC SAMPLES (IN %)

Depth (cm)	Sample Size	Min	Max	Mean	SD	CV (%)
0-15	30	0.582	1.273	0.866	0.152	17.53
15-30		0.368	0.680	0.551	0.089	16.16
30-60		0.330	0.458	0.394	0.034	8.786
60-90		0.283	0.453	0.346	0.038	11.14
90-120		0.214	0.480	0.275	0.055	20.32
120-150		0.166	0.443	0.257	0.057	22.32

III. METHODS

A. Workflow

The workflow is visualized in Fig. 5.

B. Spectral Transformation

1) Wavelet Transformation: In our study, we employed both discrete and continuous wavelet transformations (CWTs) to analyze the spectral data. Although wavelet transform has limitations in characterizing local time-domain features, it is effective for decomposing signals into frequency components, which is essential for analyzing spectral features related to SOC. Despite the complexities of hyperspectral data, the wavelet transform's balance of time and frequency resolution

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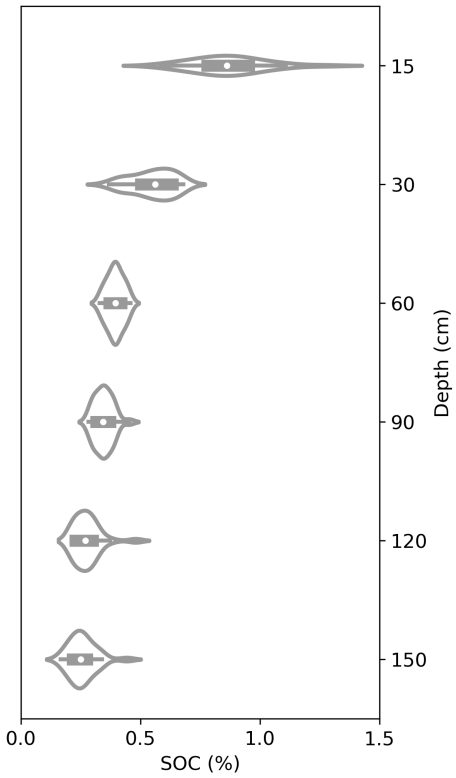


Fig. 4. SOC percentage by depth.

made it a suitable and practical tool for our objectives. Discrete wavelet transformation (DWT) decomposes signals into wavelet coefficients at various scales and positions, providing insights into the signal's frequency composition by recursively filtering the spectrum with high-pass and low-pass filters derived from the selected wavelet function [23]. This process yields detail coefficients representing high-frequency details and approximation coefficients representing the low-frequency trend within the spectrum [24]. On the other hand, CWT offers a comprehensive analysis of hyperspectral data, capturing both broad and localized spectral features that indicate variations in SOC content. CWT is favored for its ability to provide a continuous spectrum of scales for feature extraction, enhancing estimation capabilities in soil properties [25]. Hyperspectral imaging was conducted in a controlled indoor environment using HySpex VNIR and SWIR imaging cameras, minimizing atmospheric noise, and ensuring clean, accurate data. Basic denoising techniques were also applied during preprocessing to further enhance data quality. A Python package named "PyWavelets" was used to transform the spectral data [26]. We selected wavelet basis functions by testing several common types and referencing previous studies to ensure they effectively captured the spectral characteristics of the soil. In our study, we employ the following mother wavelets.

a) Biorthogonal wavelet (bior 1.3): The biorthogonal 1.3 wavelet ("bior1.3") is chosen for its precise symmetry and additional vanishing moments, which are critical for accurate feature extraction in hyperspectral data analysis [27]. Its robust edge-preserving capability ensures the integrity of nuanced spectral variations, enhancing the accuracy of SOC estimation by effectively isolating both approximation and detail coefficients. The transformation can be expressed as follows:

$$A[n] = \sum_k h_k \cdot x[2n-k] \tag{1}$$

$$D[n] = \sum_k g_k \cdot x[2n-k] \tag{2}$$

where, $A[n]$ are the approximation coefficients, $D[n]$ are the detail coefficients, h_k are the low-pass decomposition filter coefficients, g_k are the high-pass decomposition filter coefficients, and $x[n]$ is the input signal.

b) Daubechies wavelet (db2): Daubechies wavelets, particularly "db2," are valued for their compact support and significant vanishing moments, which are essential for processing complex signals like hyperspectral data [28]. The "db2" wavelet effectively captures sharp spectral discontinuities, making it ideal for SOC estimation by decomposing signals into approximation and detail components, thereby enhancing the detection of key spectral signatures related to SOC [29]. The transformation can be expressed as follows:

$$A[n] = \sum_k h_k^{\text{db2}} \cdot x[2n-k] \tag{3}$$

$$D[n] = \sum_k g_k^{\text{db2}} \cdot x[2n-k] \tag{4}$$

where, $A[n]$ are the approximation coefficients; $D[n]$ are the detail coefficients; h_k^{db2} are the low-pass decomposition filter coefficients; g_k^{db2} are the high-pass decomposition filter coefficients; and $x[n]$ is the input signal.

c) Morlet wavelet (morl): The Morlet wavelet, characterized by a cosine wave modulated by a Gaussian envelope [30], is especially effective for identifying spectral features that vary gradually across different frequencies [31]. This wavelet is well-suited for hyperspectral data analysis due to its ability to balance time and frequency resolution, making it ideal for detecting subtle changes in soil properties reflected in the spectral signatures. This decomposition can be expressed as follows:

$$\psi(t) = \exp^{-\frac{t^2}{2}} \cos(5t) \tag{5}$$

where, $\psi(t)$ represents the wavelet function, t is the time or position parameter, and exp represents the base of the natural logarithm.

d) Complex Morlet wavelets (cmor 1.5–0.5): The complex Morlet wavelet, defined by its adjustable bandwidth and center frequency, is crucial for spectral analysis in soil carbon estimation [32]. The bandwidth controls frequency resolution and feature localization, while the center frequency targets specific soil-related frequencies, optimizing spectral feature detection [33]. The center frequency parameter allows for tuning the wavelet to target frequencies associated with soil characteristics, enhancing the detection of relevant spectral features [34]. Proper tuning of these parameters balances spectral detail with spatial precision, making the complex Morlet wavelet a powerful tool for accurately extracting and analyzing soil carbon spectral information. This decomposition

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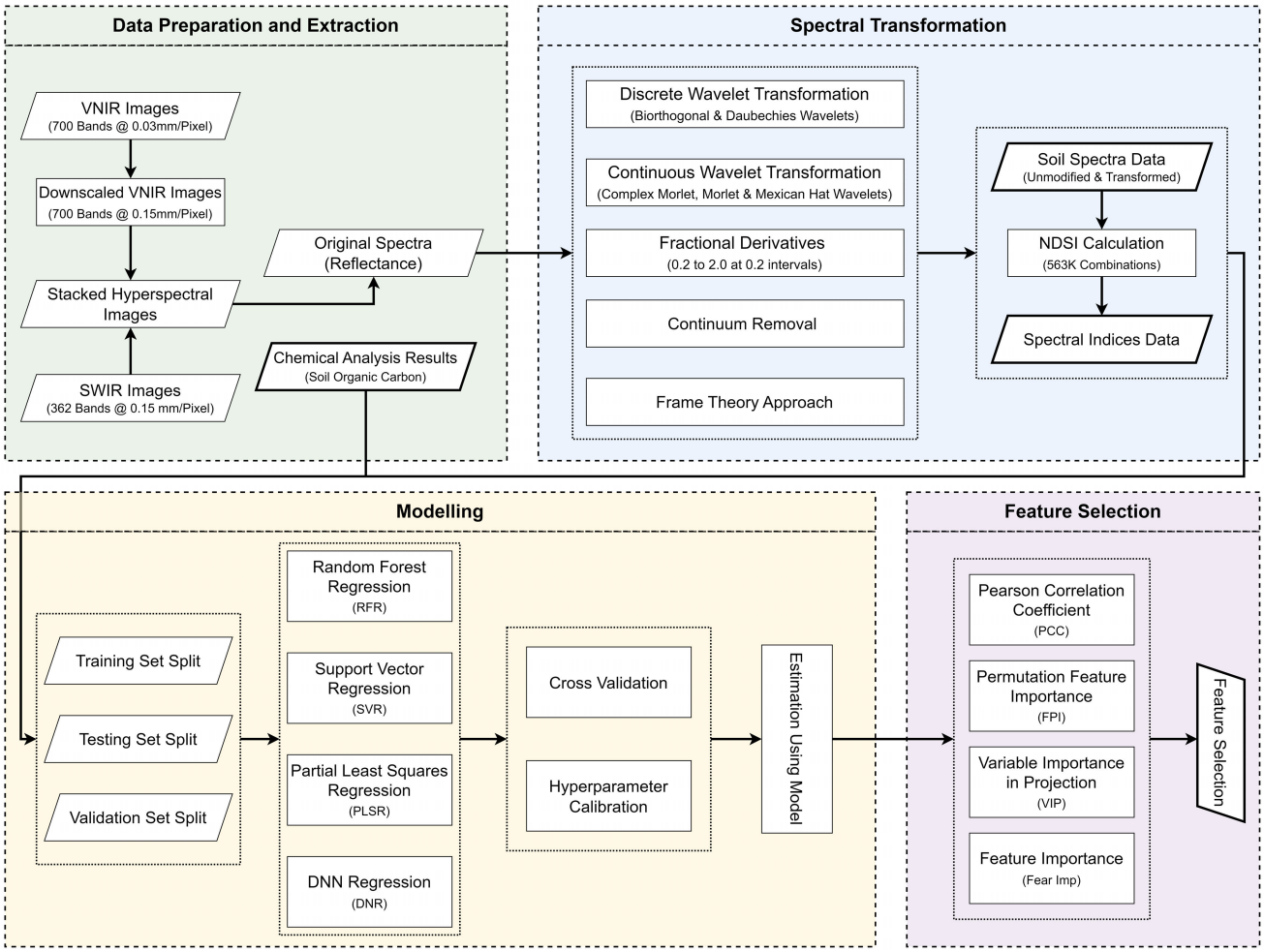


Fig. 5. Overall workflow of the data preparation to modeling pipeline.

can be expressed as follows:

$$\psi(t) = \frac{1}{\sqrt{\pi f_b}} e^{2i\pi f_c t} e^{-t^2/f_b} \tag{6}$$

where, $\psi(t)$ represents the wavelet function; f_b denotes the bandwidth parameter; f_c is the center frequency; and t is the time parameter.

e) Mexican hat wavelet (mexh): The Mexican hat wavelet, or the Ricker wavelet, is a second derivative of a Gaussian function [35], producing a waveform that is adept at highlighting rapid changes in the signal [36]. Its shape makes it particularly useful for detecting inflection points and boundaries within the spectral data [37], which can signify transitions between different soil constituents or layers. This decomposition can be expressed as follows:

$$\psi(t) = \frac{2}{\sqrt{3}\sqrt[4]{\pi}} \exp^{-\frac{t^2}{2}} \left(1 - t^2\right) \tag{7}$$

where, $\psi(t)$ represents the wavelet function, t is a nondimensional time parameter, and exp represents the base of the natural logarithm.

We used the standard settings of PyWavelets for signal decomposition, which effectively handled energy distribution during wavelet packet transformation.

2) Fractional Derivatives: Fractional derivatives, extending differentiation to noninteger orders, offer a powerful tool for enhancing feature extraction in hyperspectral imagery analysis for soil carbon estimation [38]. By emphasizing subtle spectral changes, fractional derivatives highlight variations in soil properties across spectral bands, improving SOC estimation accuracy. The derivative order controls signal smoothness, allowing adjustment of sensitivity to specific spectral features [39]. In this study, we employ the Grunwald–Letnikov (G–L) definition due to its straightforward coefficients [40]. The G–L definition is expressed as follows:

$$d^{\alpha} f(x) = \lim_{h \to 0} \frac{1}{h^{\alpha}} \sum_{m=0}^{(t-a)/h} (-1)^m \frac{\Gamma(\alpha + 1)}{m! \Gamma(\alpha - m + 1)} f(x - mh) \tag{8}$$

where, α is the order; h is the step size; t is the upper limit of fractional derivatives; a is the lower limit of fractional derivatives; and Γ is the gamma function.

3) Continuum Removal: Continuum removal in spectral analysis normalizes the spectrum against a convex hull-based continuum curve, isolating absorption features critical for identifying soil properties linked to organic carbon [41]. By interpolating between reflectance maxima, the technique

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emphasizes the depth and shape of absorption features, enhancing the accuracy and comparability of spectral data across different samples by mitigating effects like lighting variations [42]. This transformation can be expressed as follows:

$$R_{\mathrm{cr}}(\lambda) = \frac{R(\lambda)}{C(\lambda)} \tag{9}$$

where, $R_{\mathrm{cr}}(\lambda)$ is the continuum-removed reflectance at wavelength λ; $R(\lambda)$ is the original reflectance at wavelength λ; and $C(\lambda)$ is the continuum curve at wavelength λ obtained by interpolating the reflectance values at the local maxima.

4) Frame Theory Applications: Frame theory offers a mathematically rich and flexible framework for signal analysis, extending the capabilities of traditional basis-oriented approaches [43]. A frame for a Hilbert space is a collection of vectors that satisfy a specific frame condition, allowing for redundancy and enhanced representational power [44]. For our study, we employ Gabor frames due to their suitability for pinpointing localized changes within signals [45]. A Gabor system in $L^2(R)$ has the form $\{e^{2\pi imbx} g(x - na)\}_{m,n\in Z}$ for some parameters $a, b > 0$ and for some function $g \in L^2(R)$. Using the translation operators and the modulation operators, we can denote a Gabor system by $\{g_{m,n}\} = \{M_{mb} T_{na} g\}$ [46]. This construction results in frame elements that are well-localized in both the time and frequency domains. The core of a Gabor function is defined as follows [47]:

$$g(t) = e^{-\pi\beta t^2} \cos(2\pi\omega t) \tag{10}$$

where t represents a time point, ω is the frequency of the oscillation within the Gabor function, and β is the bandwidth of the Gabor function, determining its width. To further enhance the ability to capture subtle spectral nuances, we adopt an adaptive bandwidth strategy [48]. The bandwidth of each Gabor function is dynamically adjusted based on its frequency, as expressed in the following equation:

$$\beta(\omega, \omega_{\min}, \omega_{\max}) = 1.0 + \kappa \times ln\left(1 + \frac{\omega - \omega_{\min}}{\omega_{\max} - \omega_{\min}}\right) \tag{11}$$

where, ω denotes the frequency, κ is the scaling factor, and β is the bandwidth parameter. Compositional variations in soil, reflective of differing SOC content, manifest as localized features across the spectrum [49]. The adaptive Gabor frame enables the extraction of these features with the precision necessary for accurate SOC estimation. Higher frequencies often necessitate a broader bandwidth to accurately represent rapid oscillations within the signal. In contrast, lower frequencies can be effectively analyzed with a narrower bandwidth [50]. Our simplified Gabor frame construction focuses on this frequency-dependent modulation as expressed in the following equation:

$$g_\omega(t) = e^{-\pi\beta(\omega)t^2} \cos(2\pi\omega t) \tag{12}$$

where, ω denotes the frequency, $g_\omega(t)$ represents a single frame element indexed by frequency, and $\beta(\omega)$ is the bandwidth of a Gabor element, calculated based on its frequency using the adaptive bandwidth function. By tailoring the analysis to the spectral characteristics in this manner, we achieve a representation exceptionally sensitive to the subtle shifts in the soil spectra, directly correlated with SOC variations. Frame theoretic analysis was performed to extract the spectral characteristics encoded within the adaptive Gabor frame. For a fixed vector $g \in L^2(\mathbb{R})$ and $a, b > 0$, the associated Gabor frame or Weyl-Heisenberg frame G(a, b, g) is a sequence $g_{m,n}$ for which there exist two constants $0 < A < B < \infty$ such that the frame inequality is satisfied [51]. This is expressed in the following equation:

$$A \parallel f \parallel^2 \leq \sum_{m,n\in Z} |\langle f, g_{m,n}\rangle|^2 \leq B \parallel f \parallel^2, \quad f \in L^2(\mathbb{R}). \tag{13}$$

An analysis operator for a Gabor frame is a linear map defined as T_g which takes f to $\{\langle f, g_{m,n}\rangle\}_{m,n\in Z}$ and a synthesis operator T_g^* which is the adjoint of the analysis operator, takes elements $\{c_{m,n}\} \in \ell^2(Z)$ to $\sum_{m,n\in Z} c_{m,n} g_{m,n}$. The Gabor frame operator $S_g = T_g^* T_g$ is defined by the following equation:

$$S_g f = \sum_{m,n\in\mathbb{Z}} \langle f, g_{m,n}\rangle g_{m,n}. \tag{14}$$

This process mathematically decomposes the soil spectrum f into a combination of the frame elements ϕ_i as expressed in the following equation [52]:

$$Sf = \sum_{i\in I} \langle f, \phi_i\rangle \phi_i. \tag{15}$$

Where the S is the frame operator and frame coefficient $\langle f, \phi_i\rangle$ quantify how strongly the spectrum aligns with each frame element. In Python, we defined custom functions to construct individual Gabor functions and assemble the complete Gabor frame. To dynamically adjust the bandwidth of each Gabor function based on its frequency, the adaptive bandwidth strategy was incorporated. Frame analysis is performed using NumPy's [53] matrix-vector multiplication capabilities, projecting the spectral data onto the adaptive Gabor frame.

C. Modeling

In this study, an automated modeling pipeline (visualized in Fig. 5) was deployed to apply machine learning regression techniques across spectrally transformed datasets for the assessment of SOC. These models are: random forest regressor (RFR), support vector regression (SVR), partial least squares regression (PLSR), and deep neural network (DNN) regression, for their application in SOC estimation from spectral data. The implementation of these models was facilitated by the Scikit-learn package in Python [54]. All spectral features were included in the model inputs and Pearson correlation coefficient (PCC) analysis was performed to identify the most relevant spectral regions related to SOC, which provided a basis for understanding the spectral regions affecting SOC rather than refining the models' inputs. Hyperparameter tuning for the models, including RFR, SVR, and PLSR was conducted using Scikit-learn [54] and grid search functionality, while DNN tuning was facilitated by the Hyperband package [55]. A ten-fold cross-validation process was integrated within the hyperparameter tuning phase for all models.

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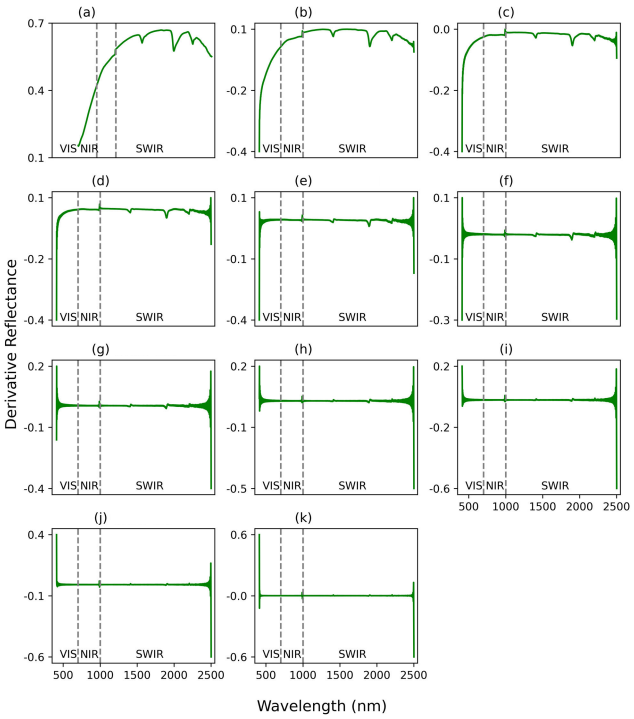


Fig. 6. Transformation of soil reflectance spectra with increasing fractional derivative orders (0.2–2.0 at 0.2 intervals). (a) Mean spectra. (b) Order 0.2. (c) Order 0.4. (d) Order 0.6. (e) Order 0.8. (f) Order 1.0. (g) Order 1.2. (h) Order 1.4. (i) Order 1.6. (j) Order 1.8. (k) Order 2.0.

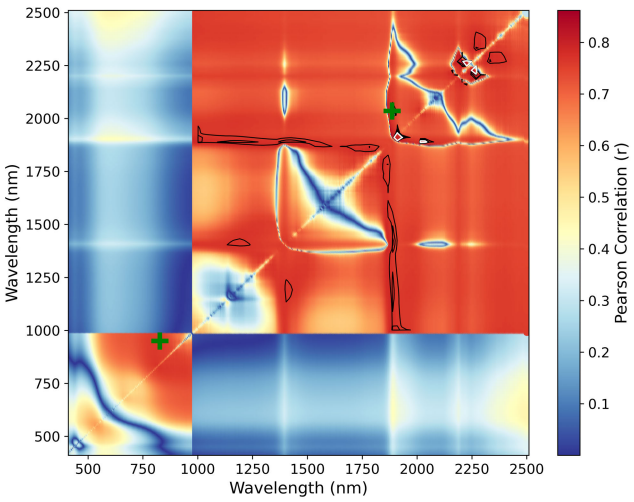


Fig. 7. Heat map of the absolute value of PCC between NDSI and SOC values. The green "+" symbols indicate the NDSI with the highest PCC in each region. The black contour lines indicate PCC from 0.75 to 0.8 while the white contour lines indicate PCC above 0.8.

IV. Results

A. Spectral Sensitivity and Feature Importance

Fractional derivative analysis significantly transformed soil reflectance spectra, revealing subtle variations indicative of soil carbon content (Fig. 6). As derivative orders increased, more pronounced spectral peaks and troughs emerged, with orders 0.4 and 1.6 [Fig. 6(c) and (i)] showing significant differences in maximum and minimum derivative values, suggesting increased sensitivity to spectral features linked to soil carbon. Normalized difference spectral indices analysis highlighted

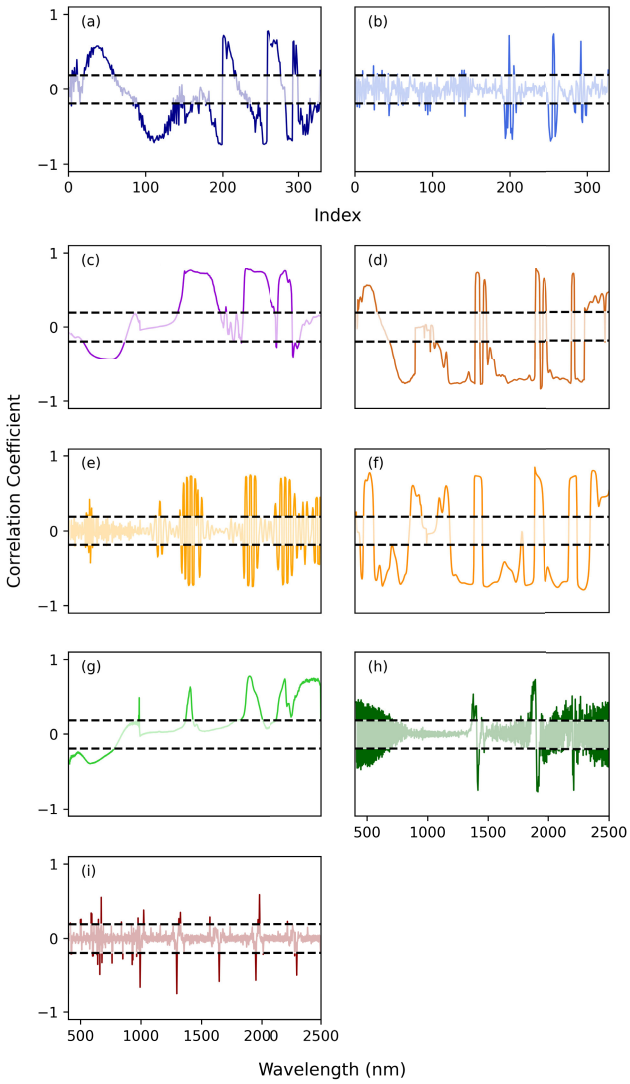


Fig. 8. PCCs between transformed spectra and SOC. Correlations outside dashed lines are significant ($p < 0.01$). (a) DWT-bior 1.3. (b) DWT-db2. (c) NT-CR. (d) CWT-cmor 1.5–0.5. (e) CWT-morl. (f) CWT-mexh. (g) FD-0.4. (h) FD-1.6. (i) GFT-adaptive.

strong correlations between SOC and specific spectral regions within the SWIR and VNIR bands (Fig. 7). The SWIR region showed the highest correlations, particularly around 1900–2100 nm and 2200–2400 nm, while the VNIR region exhibited a significant correlation band around 800–900 nm.

Feature importance analysis (Figs. 8 and 9) further underscored the significance of these spectral ranges. PCC scores (Fig. 8) indicated that certain transformations, especially CWT-mexh, emphasized critical regions in both VNIR and SWIR bands. In Fig. 9, the feature importance subplot revealed significant concentrations of important features around 1300–1500 nm, 1830–1950 nm, and 2150–2300 nm, primarily within the SWIR region. The variable importance in the projection (VIP) subplot highlighted a broader range of important features, spanning both VNIR (400–1000 nm) and SWIR regions. The feature permutation importance (FPI) subplot displayed fewer bands of high importance but consistently identified the 1850–2000 nm region as significant across all spectra. Overall, the 1850–2000 nm region within the SWIR

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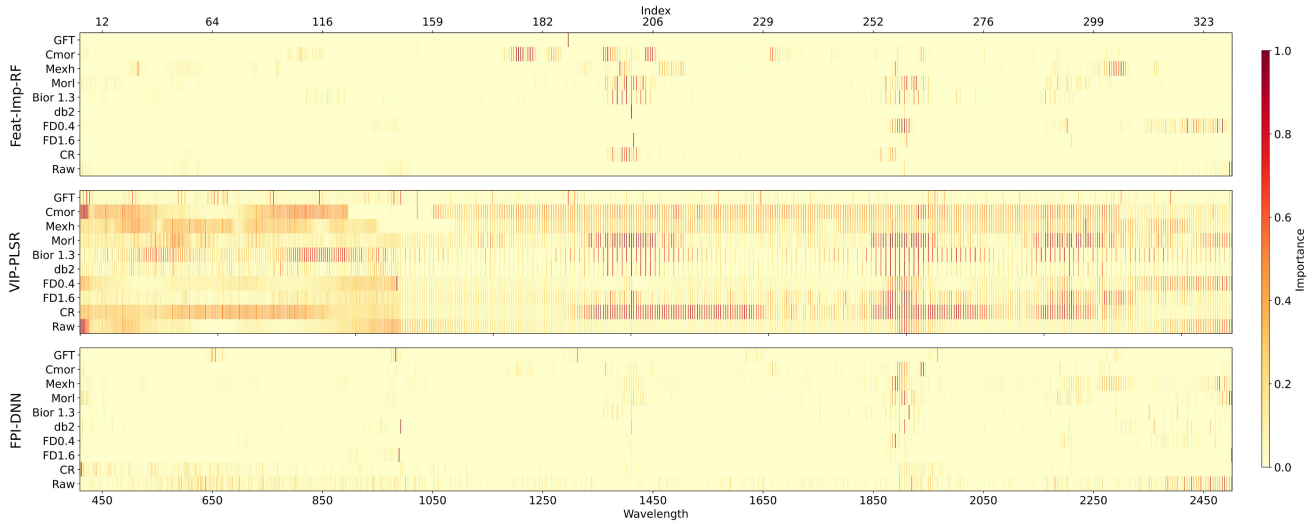


Fig. 9. Spectral feature importance for SOC estimation.

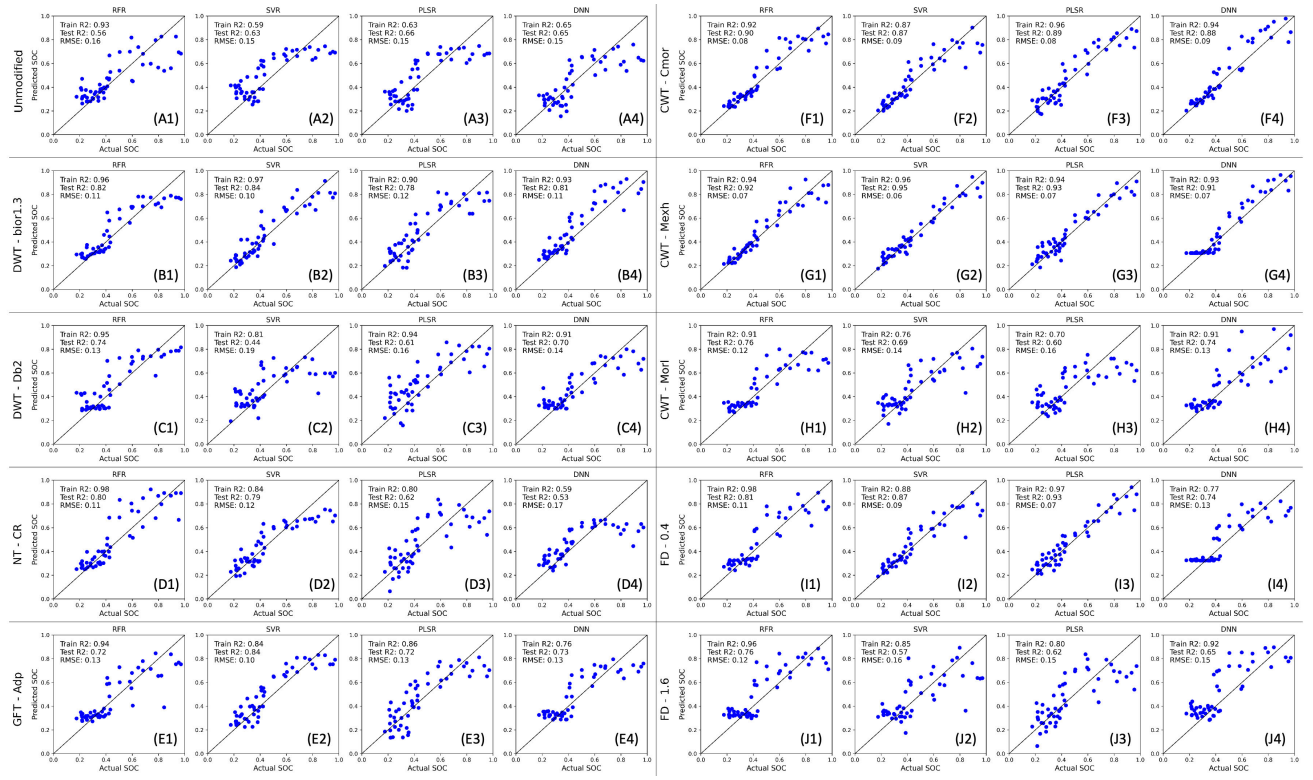


Fig. 10. Scatter plot showing the actual versus predicted SOC for each of the pipelines. The first character (A–J) indicates the spectral transformation used and the second character (1–4) indicates the ML model for the estimation.

demonstrated notable importance across all feature importance methods and spectral transformations. A smaller subset of data was used for comparing different wavelet transformations for SOC estimation [56].

B. Model Results for SOC Estimation

Model performance varied significantly depending on the modeling algorithm and spectral transformation (Fig. 10). RFR models consistently outperformed others, particularly with CWT-mexh, CWT-cmor, FD-0.4, and DWT-bior1.3, achieving test R^2 values of 0.8–0.92 and RMSE between 0.07 and 0.11. These transformations demonstrated strong potential for SOC estimation, though some models, especially RFR and PLSR, showed overfitting. PLSR models performed well, with test R^2 up to 0.93, while SVR results were more variable. DNN models, although better than SVR with unmodified spectra, generally performed similarly or slightly worse with transformed data. Transformations like CWT-mexh, with a high density of significant features, were linked to better model robustness, whereas gabor frame transformation (GFT)-adaptive, with fewer important features, led to more frequent overfitting.

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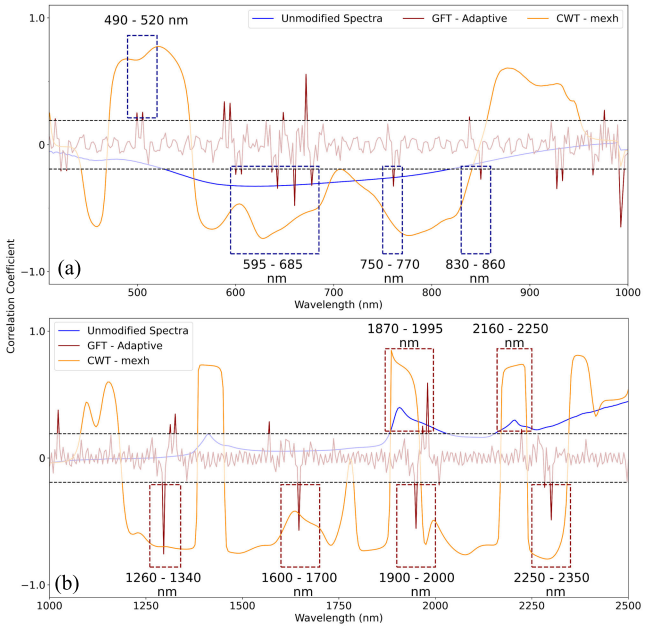


Fig. 11. Key spectral regions based on the correlation between SOC and the spectra (a) in VNIR region (b) SWIR region. Dashed lines indicate significance at $p < 0.01$.

V. Discussion

A. Key Spectral Regions for SOC Estimation

PCC analysis identified several key spectral regions with significant correlations to SOC content across the VNIR and SWIR spectra (Fig. 11). In the VNIR, positive correlations in the 490–520 nm range, influenced by humic substances, and negative correlations in the 595–685 nm, 750–770 nm, and 830–860 nm bands, related to iron oxides and moisture interactions, were observed. The SWIR revealed robust SOC-sensitive regions, notably at 1870–1995 nm and 2160–2250 nm, linked to O–H and C–H bonds in organic matter [57]. Negative correlations in bands like 1260–1340 nm and 1600–1700 nm suggest indirect SOC reflections through water and clay minerals, with the 2200–2400 nm range also showing sensitivity to organic molecules [58]. The consistent identification of these regions across both GFT-adaptive and CWT-mexh transformations underscores their robustness as SOC indicators.

Further analysis emphasized the significance of specific spectral regions for SOC estimation across various transformation methods (Fig. 12). DWT methods highlighted the NIR region, particularly around 910–912 nm, while NT-CR and CWT techniques underscored the importance of the SWIR, especially between 1865 and 2382 nm. Fractional derivatives revealed sensitivity across both VNIR and SWIR, with notable correlations from 400 to 800 nm and 2200 to 2400 nm. GFT-adaptive pinpointed unique wavelengths, such as 1296 and 1980 nm, suggesting the potential of subtle spectral features for SOC analysis, with further investigation warranted to understand these findings.

Overall, feature importance analysis (Fig. 13) demonstrated that significant SOC-related features were predominantly located in the SWIR region (67%), particularly in the 1800–2000 nm range, aligning with known absorption features

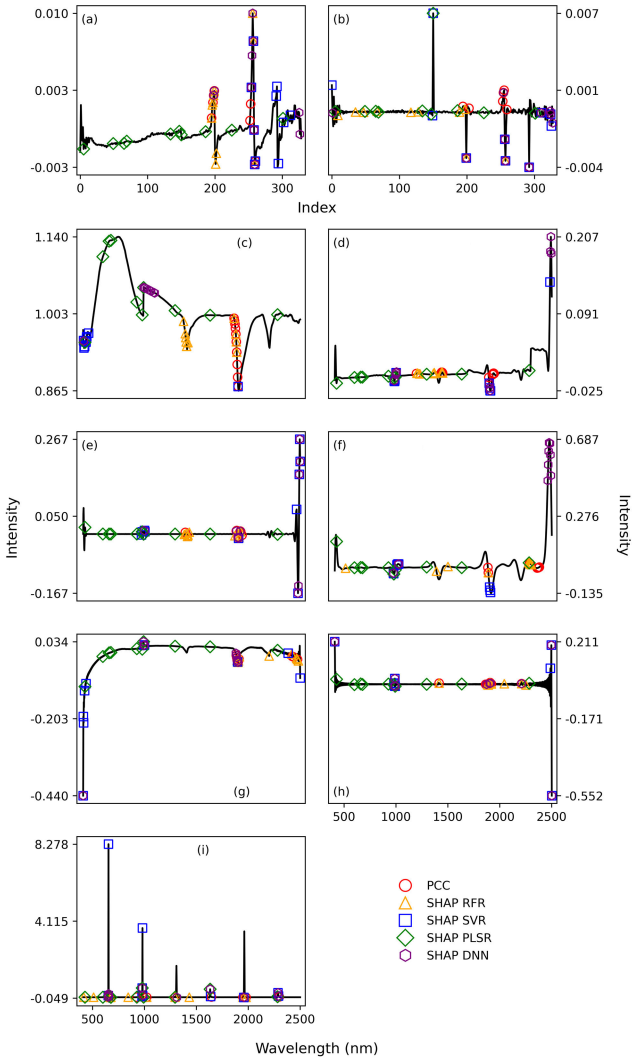


Fig. 12. Feature importances across all models and transformation techniques. Markers indicate significant features based on PCC and SHAP values. (a) DWT-bior 1.3. (b) DWT-db2. (c) NT-CR. (d) CWT-cmor 1.5–0.5. (e) CWT-morl. (f) CWT-mexh. (g) FD-0.4. (h) FD-1.6. (i) GFT-adaptive.

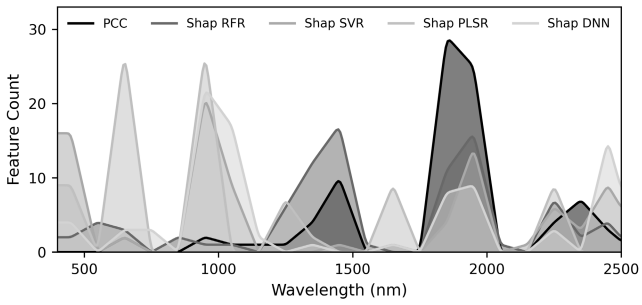


Fig. 13. Distribution of important features for estimating SOC in the VNIR-SWIR spectra.

linked to organic matter, moisture, and clay minerals [59]. The VNIR also contributed essential features, particularly in the 900–1000 nm and 600–700 nm ranges. This comprehensive analysis reaffirms the critical role of both VNIR and SWIR regions in SOC estimation and highlights underexplored spectral variations that could advance SOC characterization.

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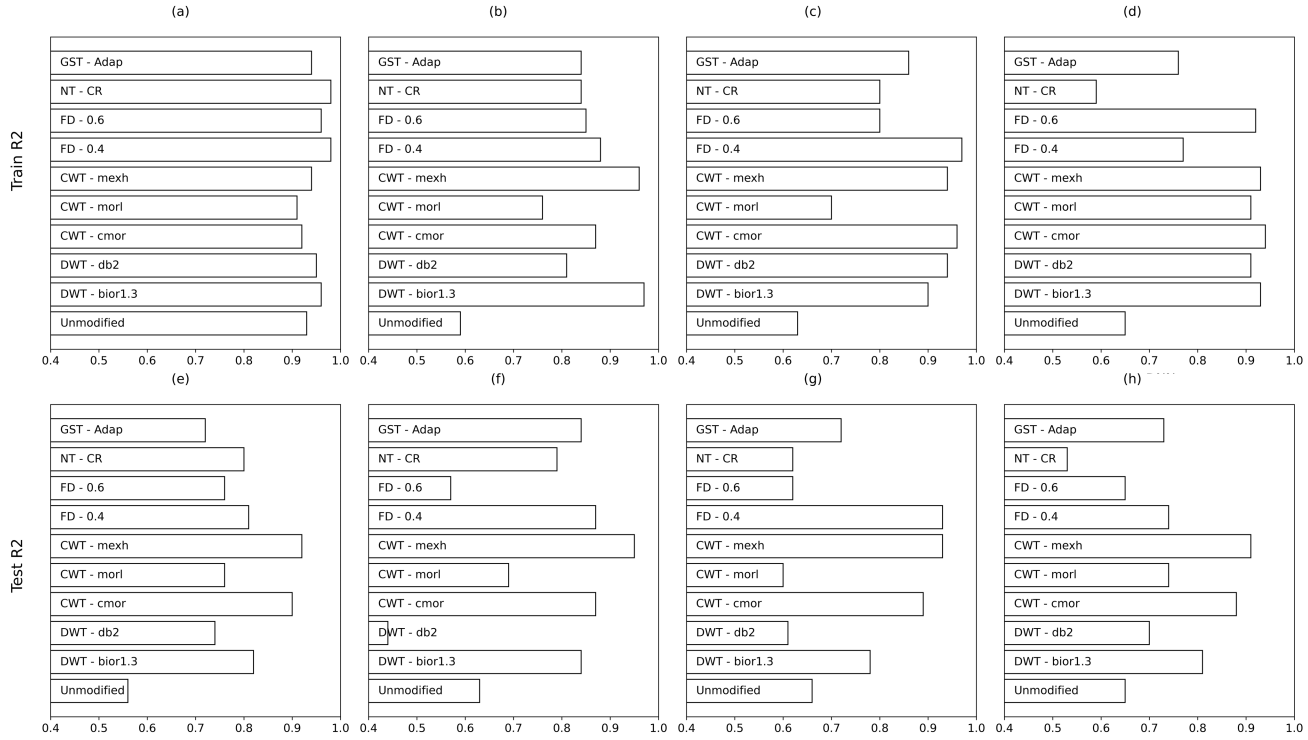


Fig. 14. Evaluation of models and spectral transformations for SOC prediction. (a)–(d) Training R^2 scores and (e)–(h) test R^2 scores for RFR, SVR, PLSR, and DNN models, respectively.

B. Model Performance Analysis

Model performance across RFR, SVR, PLSR, and DNN algorithms (Fig. 14) revealed that RFR consistently outperformed others, achieving high R^2 scores in both training and testing, with minimal overfitting. SVR and DNN models often showed larger gaps between training and testing R^2 scores, indicating higher overfitting, particularly with DWT-db2 and FD-1.6 transformations. PLSR exhibited moderate performance, with R^2 scores ranging from 0.6 to 0.78 and less overfitting. CWT-mexh, FD, and GFT-adaptive transformations significantly improved model performance, with CWT-mexh delivering the highest test R^2 scores across all models (up to 0.95 with SVR). GFT-adaptive, despite identifying fewer significant features, consistently achieved test R^2 scores above 0.7 across all algorithms, demonstrating its efficiency and potential in scenarios with limited spectral resolution, making it valuable for SOC prediction with fewer spectral bands.

VI. Conclusion

This study advances methodologies for SOC estimation by integrating hyperspectral imaging with signal processing techniques. Notably, CWT-mexh models achieved R^2 scores up to 0.95, with the RFR outperforming SVR, PLSR, and DNN models in both accuracy and reduced overfitting. Key findings include the following:

1) The SWIR region, especially 1800–2000 nm, shows the strongest correlation with SOC content (above 0.8).
2) CWT (Mexican hat and complex Morlet), fractional derivatives (order 0.4 and 1.6), and frame transformation methods significantly improve SOC prediction accuracy.
3) RFR, paired with selected spectral transformations, offers the most accurate and robust SOC estimation framework.
4) The frame transformation method's success with fewer features suggests its potential where spectral resolution is limited, while CWT (Mexh) excels with high spectral resolution.
5) Consistent spectral regions (1800–2000 nm, 900–1000 nm, and weaker correlations in 400–700 nm) warrant further investigation regarding specific SOC constituents.

This study primarily focused on laboratory-based spectroscopy and did not address depth-sensitive analysis. Future research should explore spectral responses across soil depths, scalability for airborne and spaceborne platforms, and assessments across diverse soil types to validate the robustness and generalizability of the findings. Additionally, integrating GFT-adaptive with other spectral methods may further enhance SOC prediction accuracy.

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References

[1] G. Gebeyehu, T. Soromessa, and D. Teketay, "Organic carbon stocks, dynamics and restoration in relation to soils of agroecosystems in ethiopia: A review," *Int. J. Environ.*, vol. 6, no. 1, pp. 1–22, Feb. 2017, doi: 10.3126/ije.v6i1.16864.

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[2] D. Angers et al., "A well-established fact: Rapid mineralization of organic inputs is an important factor for soil carbon sequestration," *Eur. J. Soil Sci.*, vol. 73, no. 3, May 2022, Art. no. e13242, doi: 10.1111/ejss.13242.

[3] H. Karabcová, L. Pospíšilová, K. Fiala, P. Škarpa, and M. Bjelková, "Effect of organic fertilizers on soil organic carbon and risk trace elements content in soil under permanent grassland," *Soil Water Res.*, vol. 10, no. 4, pp. 228–235, Dec. 2015, doi: 10.17221/5/2015-swr.

[4] C. Jansson et al., "Crops for carbon farming," *Frontiers Plant Sci.*, vol. 12, pp. 1–12, Jun. 2021, doi: 10.3389/fpls.2021.636709.

[5] E. Goidts, B. Van Wesemael, and M. Crucifix, "Magnitude and sources of uncertainties in soil organic carbon (SOC) stock assessments at various scales," *Eur. J. Soil Sci.*, vol. 60, no. 5, pp. 723–739, Oct. 2009, doi: 10.1111/j.1365-2389.2009.01157.x.

[6] L. Henneron et al., "Bioenergetic control of soil carbon dynamics across depth," *Nature Commun.*, vol. 13, no. 1, p. 7676, Dec. 2022, doi: 10.1038/s41467-022-34951-w.

[7] C. Zhang, Y. Tang, X. Xu, and G. Kiely, "Towards spatial geochemical modelling: Use of geographically weighted regression for mapping soil organic carbon contents in Ireland," *Appl. Geochemistry*, vol. 26, no. 7, pp. 1239–1248, Jul. 2011, doi: 10.1016/j.apgeochem.2011.04.014.

[8] E. Ben-Dor et al., "Using imaging spectroscopy to study soil properties," *Remote Sens. Environ.*, vol. 113, pp. S38–S55, Sep. 2009, doi: 10.1016/j.rse.2008.09.019.

[9] L. Guo et al., "Mapping soil organic carbon stock by hyperspectral and time-series multispectral remote sensing images in low-relief agricultural areas," *Geoderma*, vol. 398, Sep. 2021, Art. no. 115118, doi: 10.1016/j.geoderma.2021.115118.

[10] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024, doi: 10.1109/TPAMI.2024.3362475.

[11] M. Zhao, S. Shi, J. Chen, and N. Dobigeon, "A 3-D-CNN framework for hyperspectral unmixing with spectral variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521914, doi: 10.1109/TGRS.2022.3141387.

[12] Y. Ding et al., "Multi-feature fusion: Graph neural network and CNN combining for hyperspectral image classification," *Neurocomputing*, vol. 501, pp. 246–257, Aug. 2022, doi: 10.1016/j.neucom.2022.06.031.

[13] L. Xu et al., "Estimation of organic carbon in anthropogenic soil by VIS-NIR spectroscopy: Effect of variable selection," *Remote Sens.*, vol. 12, no. 20, p. 3394, Oct. 2020, doi: 10.3390/rs12203394.

[14] Y. Hong et al., "Transferability of vis-NIR models for soil organic carbon estimation between two study areas by using spiking," *Soil Sci. Soc. Amer. J.*, vol. 82, no. 5, pp. 1231–1242, Sep. 2018, doi: 10.2136/sssaj2018.03.0099.

[15] Y. Liu et al., "Transferability of a visible and near-infrared model for soil organic matter estimation in riparian landscapes," *Remote Sens.*, vol. 6, no. 5, pp. 4305–4322, May 2014, doi: 10.3390/rs6054305.

[16] K. Gröchenig, "Localization of frames, Banach frames, and the invertibility of the frame operator," *J. Fourier Anal. Appl.*, vol. 10, no. 2, pp. 105–132, Mar. 2004, doi: 10.1007/s00041-004-8007-1.

[17] P. Balazs, N. Holighaus, T. Necciari, and D. Stoeva, "Frame theory for signal processing in psychoacoustics," in *Excursions in Harmonic Analysis: The February Fourier Talks at the Norbert Wiener Center*, vol. 5, R. Balan, J. J. Benedetto, W. Czaja, M. Dellatorre, and K. A. Okoudjou, Eds., Cham, Switzerland: Springer, 2017, pp. 225–268.

[18] J. Kovačević and A. Chebira, "An introduction to frames," *Found. Trends Signal Process.*, vol. 2, no. 1, pp. 1–94, 2007, doi: 10.1561/2000000006.

[19] A. Aldroubi, A. Baskakov, and I. Krishtal, "Slanted matrices, Banach frames, and sampling," *J. Funct. Anal.*, vol. 255, no. 7, pp. 1667–1691, Oct. 2008, doi: 10.1016/j.jfa.2008.06.024.

[20] A. J. Sykes et al., "Characterising the biophysical, economic and social impacts of soil carbon sequestration as a greenhouse gas removal technology," *Global Change Biol.*, vol. 26, no. 3, pp. 1085–1108, Mar. 2020, doi: 10.1111/gcb.14844.

[21] R. P. Sishodia, R. L. Ray, and S. K. Singh, "Applications of remote sensing in precision agriculture: A review," *Remote Sens.*, vol. 12, no. 19, p. 3136, Sep. 2020, doi: 10.3390/rs12193136.

[22] K. Lenhard, A. Baumgartner, P. Gege, S. Nevas, S. Nowy, and A. Sperling, "Impact of improved calibration of a NEO HySpex VNIR-1600 sensor on remote sensing of water depth," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 11, pp. 6085–6098, Nov. 2015, doi: 10.1109/TGRS.2015.2431743.

[23] S. Saini and L. Dewan, "Application of discrete wavelet transform for analysis of genomic sequences of mycobacterium tuberculosis," *SpringerPlus*, vol. 5, no. 1, p. 64, Dec. 2016, doi: 10.1186/s40064-016-1668-9.

[24] B. K. Pandey, H. Tiwari, and D. Khare, "Trend analysis using discrete wavelet transform (DWT) for long-term precipitation (1851–2006) over India," *Hydrologi. Sci. J.*, vol. 62, no. 13, pp. 2187–2208, Oct. 2017, doi: 10.1080/02626667.2017.1371849.

[25] J. Yang, X. Li, and X. Ma, "Improving the accuracy of soil organic carbon estimation: CWT-random frog-XGBoost as a prerequisite technique for in situ hyperspectral analysis," *Remote Sens.*, vol. 15, no. 22, p. 5294, Nov. 2023, doi: 10.3390/rs15225294.

[26] G. Lee, R. Gommers, F. Waselewski, K. Wohlfahrt, and A. O'Leary, "PyWavelets: A Python package for wavelet analysis," *J. Open Source Softw.*, vol. 4, no. 36, p. 1237, Apr. 2019, doi: 10.21105/joss.01237.

[27] X. Xu, S. Chen, Z. Xu, Y. Yu, S. Zhang, and R. Dai, "Exploring appropriate preprocessing techniques for hyperspectral soil organic matter content estimation in black soil area," *Remote Sens.*, vol. 12, no. 22, p. 3765, Nov. 2020, doi: 10.3390/rs12223765.

[28] T. Guo, T. Zhang, E. Lim, M. López-Benítez, F. Ma, and L. Yu, "A review of wavelet analysis and its applications: Challenges and opportunities," *IEEE Access*, vol. 10, pp. 58869–58903, 2022, doi: 10.1109/ACCESS.2022.3179517.

[29] L. Shen et al., "Hyperspectral estimation of soil organic matter content using different spectral preprocessing techniques and PLSR method," *Remote Sens.*, vol. 12, no. 7, p. 1206, Apr. 2020, doi: 10.3390/rs12071206.

[30] L. P. A. Arts and E. L. van den Broek, "The fast continuous wavelet transformation (fCWT) for real-time, high-quality, noise-resistant time-frequency analysis," *Nature Comput. Sci.*, vol. 2, no. 1, pp. 47–58, Jan. 2022, doi: 10.1038/s43588-021-00183-z.

[31] S. Kaewpijit, J. Le Moigne, and T. El-Ghazawi, "Automatic reduction of hyperspectral imagery using wavelet spectral analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 4, pp. 863–871, Apr. 2003, doi: 10.1109/TGRS.2003.810712.

[32] W. Shao, J. Barras, and P. Kosmas, "A novel wavelets method for cancelling time-varying interference in NQR signal detection," *Signal Process.*, vol. 154, pp. 238–249, Jan. 2019, doi: 10.1016/j.sigpro.2018.08.016.

[33] S. J. Ward, R. Layouni, S. Arshavsky-Graham, E. Segal, and S. M. Weiss, "Morlet wavelet filtering and phase analysis to reduce the limit of detection for thin film optical biosensors," *ACS Sensors*, vol. 6, no. 8, pp. 2967–2978, Aug. 2021, doi: 10.1021/acssensors.1c00787.

[34] A. Biswas, H. P. Cresswell, R. A. V. Rossel, and B. C. Si, "Characterizing scale- and location-specific variation in non-linear soil systems using the wavelet transform," *Eur. J. Soil Sci.*, vol. 64, no. 5, pp. 706–715, Oct. 2013, doi: 10.1111/ejss.12063.

[35] H. M. Srivastava, A. Singh, A. Rawat, and S. Singh, "A family of Mexican hat wavelet transforms associated with an isometry in the heat equation," *Math. Methods Appl. Sci.*, vol. 44, no. 14, pp. 11340–11349, Sep. 2021, doi: 10.1002/mma.7492.

[36] S. Lovejoy and D. Schertzer, "Haar wavelets, fluctuations and structure functions: Convenient choices for geophysics," *Nonlinear Processes Geophys.*, vol. 19, no. 5, pp. 513–527, Sep. 2012, doi: 10.5194/npg-19-513-2012.

[37] P. Vielva, "A comprehensive overview of the cold spot," *Adv. Astron.*, vol. 2010, no. 1, Jan. 2010, Art. no. 592094, doi: 10.1155/2010/592094.

[38] Y.-F. Pu, "Fractional-order Euler–Lagrange equation for Fractional-order variational method: A necessary condition for fractional-order fixed boundary optimization problems in signal processing and image processing," *IEEE Access*, vol. 4, pp. 10110–10135, 2016, doi: 10.1109/ACCESS.2016.2636159.

[39] J. Wang, J. Ding, A. Abulimiti, and L. Cai, "Quantitative estimation of soil salinity by means of different modeling methods and visible-near infrared (VIS–NIR) spectroscopy, Ebinur lake Wetland, Northwest China," *PeerJ*, vol. 6, p. e4703, May 2018, doi: 10.7717/peerj.4703.

[40] J. Guan, J. Ou, Z. Lai, and Y. Lai, "Medical image enhancement method based on the fractional order derivative and the directional derivative," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 32, no. 3, Mar. 2018, Art. no. 1857001, doi: 10.1142/s021800141857001x.

[41] M. F. Noomen, A. K. Skidmore, F. D. van der Meer, and H. H. T. Prins, "Continuum removed band depth analysis for detecting the effects of natural gas, methane and ethane on maize reflectance," *Remote Sens. Environ.*, vol. 105, no. 3, pp. 262–270, Dec. 2006, doi: 10.1016/j.rse.2006.07.009.

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[42] M. Loum, M. Diack, N. Y. B. Ndour, and D. Masse, "Effect of the continuum removal in predicting soil organic carbon with near infrared spectroscopy (NIRS) in the Senegal sahelian soils," *Open J. Soil Sci.*, vol. 6, no. 9, pp. 135–148, 2016, doi: 10.4236/ojss.2016.69014.

[43] W.-L. Hwang, P.-T. Huang, B.-C. Kung, J. Ho, and T.-L. Jong, "Frame-based sparse analysis and synthesis signal representations and Parseval K-SVD," *IEEE Trans. Signal Process.*, vol. 67, no. 12, pp. 3330–3343, Jun. 2019, doi: 10.1109/TSP.2019.2916105.

[44] P. G. Casazza and R. G. Lynch, "A brief introduction to Hilbert space frame theory and its applications," 2015, *arXiv:1509.07347*.

[45] H. G. Feichtinger and T. Strohmer, *Gabor Analysis and Algorithms: Theory and Applications*. Boston, MA, USA: Birkhäuser, 2012.

[46] F. Bergeaud and S. Mallat, "Matching pursuit of images," in *Wavelet Analysis and Its Applications*, vol. 7, Y. Zeevi and R. Coifman, Eds., New York, NY, USA: Academic, 1998, pp. 285–300.

[47] Y. Belov, A. Kulikov, and Y. Lyubarskii, "Gabor frames for rational functions," *Inventiones Mathematicae*, vol. 231, no. 2, pp. 431–466, Feb. 2023, doi: 10.1007/s00222-022-01151-8.

[48] P. Balazs, M. Dörfler, F. Jaillet, N. Holighaus, and G. Velasco, "Theory, implementation and applications of nonstationary Gabor frames," *J. Comput. Appl. Math.*, vol. 236, no. 6, pp. 1481–1496, Oct. 2011, doi: 10.1016/j.cam.2011.09.011.

[49] C. Rasmussen et al., "Beyond clay: Towards an improved set of variables for predicting soil organic matter content," *Biogeochemistry*, vol. 137, no. 3, pp. 297–306, Feb. 2018, doi: 10.1007/s10533-018-0424-3.

[50] V. V. Moca, H. Bârzan, A. Nagy-Dăbâcan, and R. C. Mureşan, "Time-frequency super-resolution with superlets," *Nature Commun.*, vol. 12, no. 1, p. 337, Jan. 2021, doi: 10.1038/s41467-020-20539-9.

[51] A. J. E. M. Janssen, "The duality condition for Weyl-Heisenberg frames," in *Gabor Analysis and Algorithms: Theory and Applications*, H. G. Feichtinger and T. Strohmer, Eds., Boston, MA, USA: Birkhäuser, 1998, pp. 33–84.

[52] M. Dörfler, "Quilted Gabor frames—A new concept for adaptive time-frequency representation," *Adv. Appl. Math.*, vol. 47, no. 4, pp. 668–687, Oct. 2011, doi: 10.1016/j.aam.2011.02.007.

[53] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2.

[54] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Nov. 2011. [Online]. Available: http://jmlr.org/papers/v12/pedregosa11a.html

[55] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, "Hyperband: A novel bandit-based approach to hyperparameter optimization," *J. Mach. Learn. Res.*, vol. 18, no. 185, pp. 1–52, 2018.

[56] B. Roy, V. Sagan, H. Alifu, J. Saxton, and N. Shakoor, "Comparative analysis of wavelet transformation techniques in enhancing soil organic carbon detection through hyperspectral imaging," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 11, Jul. 2024, pp. 8010–8014, doi: 10.1109/igarss53475.2024.10641176.

[57] B. Koirala, Z. Zahiri, and P. Scheunders, "A robust supervised method for estimating soil moisture content from spectral reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539813, doi: 10.1109/TGRS.2022.3212600.

[58] T. Shi, Y. Chen, H. Liu, J. Wang, and G. Wu, "Soil organic carbon content estimation with laboratory-based visible–near-infrared reflectance spectroscopy: Feature selection," *Appl. Spectrosc.*, vol. 68, no. 8, pp. 831–837, Aug. 2014, doi: 10.1366/13-07294.

[59] A. Bayer, M. Bachmann, A. Müller, and H. Kaufmann, "A comparison of feature-based MLR and PLS regression techniques for the prediction of three soil constituents in a degraded South African ecosystem," *Appl. Environ. Soil Sci.*, vol. 2012, pp. 1–20, Sep. 2012, doi: 10.1155/2012/971252.

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